

■ Part A: Conceptual Foundation (Theory)

Machine Learning — Regularization, Cross-Validation & Tree-Based Models

Instruction: Answer briefly (2–4 lines each), following the terminology and organization shown in the provided task.

1. What is Regularization in Machine Learning? Why is it needed?

Regularization is a technique used to control model complexity by adding a penalty for large model coefficients to the loss function. It helps reduce overfitting, improves generalization on unseen data, and makes the model less sensitive to noise in the training dataset.

2. Difference between Ridge Regression (L2) and Lasso Regression (L1).

Ridge Regression uses an L2 penalty, which shrinks coefficients toward zero but normally keeps all features in the model. Lasso Regression uses an L1 penalty and can shrink some coefficients exactly to zero, so it can also perform feature selection.

3. What is Cross-Validation and why is it important?

Cross-Validation is a model-evaluation technique that repeatedly divides data into training and validation portions. It provides a more reliable estimate of model performance, helps detect overfitting, and supports better model and hyperparameter selection.

4. Explain the following cross-validation techniques:

K-Fold Cross-Validation: The dataset is divided into K folds. The model trains on K–1 folds and validates on the remaining fold, repeating until every fold has been used for validation.

Stratified K-Fold Cross-Validation: Similar to K-Fold, but it keeps the class distribution approximately consistent across folds. It is especially useful for classification datasets with imbalanced classes.

Leave-One-Out Cross-Validation (LOOCV): One observation is kept for validation while all other observations are used for training. This is repeated for every observation in the dataset.

Time Series Split: Data is divided according to chronological order. Training uses earlier observations and validation uses later observations, preventing future information from being used to predict the past.

5. Why are tree-based models less sensitive to feature scaling?

Tree-based models make decisions using feature thresholds, such as whether a value is less than or greater than a split point. Because the split depends on ordering rather than distance or coefficient magnitude, changing a feature's scale usually does not change the tree's decisions.

Quick Revision Table

Concept	Key Point
Regularization	Controls model complexity and reduces overfitting.
Ridge (L2)	Shrinks coefficients; generally retains all features.

Lasso (L1)	Can shrink coefficients to zero and perform feature selection.
Cross-Validation	Provides a stronger estimate of model generalization.
Tree Models	Usually do not require feature scaling.